

MULTI-DIMENSIONAL PROSODY JUDGMENT FOR LIVE STREAMING SPEECH SYNTHESIS

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ABSTRACT

We present Live-SpeechJudge, a pairwise prosody evaluator for e-commerce live streaming speech synthesis, and Decouple-Live-SpeechJudge (D-LSJ), which enables different prosody dimensions to produce distinct judgments rather than follow one overall verdict. D-LSJ independently queries four core dimensions, masks uncertain pair-dimensions during SFT, removes the overall-verdict target, and applies each normalized GRPO advantage only to its dimension’s rationale span. Across four main human-annotated test sets, 10-sample balanced-order Live-SpeechJudge has higher point accuracy than one Gemini judgment, although it uses more inference. On four dimension-specific human test sets, span-local GRPO changes pooled 10-sample agreement from 84.10% to 86.10%; all per-dimension confidence intervals include zero. When selecting one utterance from eight TTS candidates by pairwise comparison, the selected utterance falls within the human top 1/2/3 in 72.06%/77.94%/85.29% of 136 high-confidence sets.

Index Terms— prosody evaluation, expressive speech evaluation, speech reward model, reinforcement learning

1. INTRODUCTION

Live streaming TTS requires fluency, varied intonation, appropriate emotion, high energy, sales-cue emphasis, and natural shifts between product explanation and audience interaction. Scalar naturalness scores cannot capture these requirements. We study pairwise evaluators for selecting the best of N sampled utterances and, in future work, supplying rewards for TTS preference optimization [1, 2].

Reference-free MOS predictors [3, 4, 5] are inexpensive but do not capture the interaction-mode and expressive phenomena required here. SpeechJudge-GRM [6] provides rationale-based judgments but is trained primarily for naturalness and shows substantial position bias under our protocol. Gemini-3.1-pro-preview [7] is stronger but too costly for the intended deployment setting. We therefore distill it into Qwen3-Omni [8].

A further problem is *verdict coupling*: dimension-level judgments often follow the overall winner, collapsing a nominally multi-dimensional judge to one preference bit. Our main goal is to decouple these dimensions so that fluency, intonation, emotion, and live streaming expressiveness can yield different verdicts. This work makes three contributions. First, we define a seven-dimension protocol and evaluate on 1,043 internal human-annotated pairs plus a 222-pair transfer set. Second, Live-SpeechJudge combines confidence filtering, curriculum SFT, dimension-weighted loss, and GRPO. Third, D-LSJ removes the overall verdict and localizes supervision by pair and dimension: uncertain SFT sections receive

zero loss, and each GRPO advantage updates only its corresponding rationale span.

2. RELATED WORK

MOS predictors regress naturalness or transmission-quality scores from isolated clips [3, 4, 5]. Recent benchmarks instead use audio-language models for difficult expressive judgments. EmergentTTS-Eval [9] targets prosodic and linguistic challenges, while SwanBench-Speech [10] evaluates long-form acoustics, semantics, and expressiveness. These works establish model-as-a-judge evaluation but do not build a low-cost per-utterance reward vector.

SpeechJudge [6] contributes a 99K-pair feedback corpus, SpeechJudge-Eval, and a generative reward model trained by rationale SFT and GRPO. GSRM [11] similarly uses structured acoustic reasoning. UniSRM [12] unifies several speech assessment tasks and rewards average per-dimension agreement, but broadcasts one group-level advantage over the entire completion. CEAval [13] evaluates context-conditioned expressive appropriateness. In contrast, D-LSJ removes the final-answer target and assigns an independent normalized advantage to each dimension’s tokens, preventing direct cross-dimension cancellation in the token-level objective.

3. TASK AND DATA

For two clips (a_A, a_B) rendering transcript x , the judge produces a rationale c_d and scores $(s_d^A, s_d^B) \in \{1, \dots, 10\}^2$ for dimension d . Its per-dimension verdict is

$$v_d = \text{sgn}(s_d^A - s_d^B) \in \{A, \text{Tie}, B\}. \quad (1)$$

Live-SpeechJudge evaluates four always-on dimensions—fluency and naturalness, intonation variation, emotional expression, and live streaming expressiveness—plus three transcript-activated dimensions: key-information emphasis, emotional-state switching, and interaction-mode switching. It also emits an overall verdict. D-LSJ emits only the four core dimensions and no overall conclusion; when needed, an aggregate verdict is computed externally from the score differences.

Table 1 summarizes the rubric. C1–C4 are always evaluated, whereas L1–L3 are transcript-gated; absent cues yield a non-activated section with zero scores. All experiments use the high-expressiveness profile.

Our main suite contains 1,043 pairs and covers three difficulty regimes. T1 compares qwen3-TTS base with SFT (417; T1* is its 119-pair unanimous subset). T2 compares two independent samples from qwen3-TTS-GDPO (111 unanimous pairs), testing

Table 1. Seven-dimension live streaming prosody rubric. “Text” means that the dimension is evaluated only when the transcript contains the corresponding cue.

ID	Dimension	Gate	Principal attributes
C1	Fluency and Naturalness	Always	Rate stability, phrasing, pauses, coherence
C2	Intonation Variation	Always	Pitch range, final contours, stress, sentence contrast
C3	Emotional Expression	Always	Emotion-content match, intensity, warmth and appeal
C4	Live Streaming Expressiveness	Always	Energy, rate drive, rhythm, real-host quality
L1	Key Information Emphasis	Text	Prominence of price, discount, scarcity, and calls to action
L2	Emotional State Switching	Text	Capture and smoothness of emotional turning points
L3	Interaction Mode Switching	Text	Delivery change between explanation and audience address

whether a judge can distinguish variation within one model. T3 compares qwen3-TTS-GDPO with BERT-CosyVoice [14] (315), and T4 compares human recordings with qwen3-TTS (200). T5 adds 222 IndexTTS2-versus-FireRedTTS pairs to probe transfer to a model combination absent from the coupled judge’s training.

Construction of T1–T4 is stratified rather than opportunistic. From a pool of 135 speakers, we sample 114 across warm-recommendation, passionate-promotion, and professional-explanation styles. Product categories are matched to speaker style. Transcripts combine automatic transcripts from real streams with generated copy for product entry, product explanation, and order prompting. Durations are balanced across 5–10, 10–20, and 20–30 seconds. Together, the suite covers cross-model, same-model, and human-versus-TTS comparisons while holding the transcript fixed within every pair, reducing lexical-content confounding.

Three trained contractor annotators label each main pair independently under a written protocol while blinded to system identity and audio source. They use the same rubric definitions and pair-wise criteria as the teacher prompt. Across T1–T4, 45% of final labels are unanimous 3–0 decisions and 55% contain two agreeing votes plus one abstention; objective exclusions are limited to corrupted audio, transcript mismatch, or protocol violations. We freeze every evaluation partition before testing, and no evaluation audio, pair, normalized transcript, speaker, or source stream occurs in the corresponding training or validation data. Separate human test sets for C1–C4 contain 224, 172, 142, and 160 unanimous non-tie pairs, respectively. Evaluation audio is not publicly redistributed because source recordings are subject to copyright and licensing restrictions.

4. METHOD

4.1. Swap-consistent distillation and curriculum

We query Gemini four times per training pair: twice in the presented order and twice after swapping A/B. Votes from swapped calls are mapped back to the physical clips. Live-SpeechJudge retains only 4–0 outcomes and balances presentation order so that the winner occurs equally often in slots A and B. Its two-stage curriculum first uses human-versus-TTS comparisons to teach the rubric, then uses cross-model and same-model comparisons for fine discrimination. We attach rank-32 LoRA [15] to the Qwen3-Omni attention projections and merge the adapter after SFT.

For section token set T_d and dimension weight α_d , the weighted SFT loss is

$$\mathcal{L}_{\text{SFT}} = -\frac{\sum_d \alpha_d \sum_{t \in T_d} \log \pi_\theta(y_t | y_{<t}, a_A, a_B, x)}{\sum_d \alpha_d |T_d|}. \quad (2)$$

The coupled judge uses weights (1.5, 0.7, 1.5, 1.5, 1.5, 0.7, 0.7)

over C1–C4 and L1–L3, with weights 1.0 and 2.0 on the summary and conclusion, respectively.

4.2. Dimension-wise supervision

Verdict coupling can enter through both the label and the objective. To reduce the first source, D-LSJ queries Gemini four times *per core dimension*; each call contains only that dimension’s definition and returns only its rationale and score pair. Votes from swapped calls are again restored to physical clips. We retain a pair-dimension when its four calls yield 4–0, 3–1, or three agreeing non-tie votes plus one abstention. Other pair-dimensions receive zero SFT loss, while confident dimensions from the same pair remain trainable. Thus retention occurs at the (pair, dimension) level rather than discarding a complete pair because one section is uncertain. The retained sections are concatenated into one C1–C4 rationale target with no summary or overall conclusion.

Both D-LSJ SFT stages use rank-64 QLoRA and learning rate 5×10^{-5} , for 8 and 14 epochs, respectively, with batch size 1 and eight-stage gradient accumulation.

For GRPO, $G = 8$ completions are sampled for prompt q . For any group-level signal z_i , we use

$$\mathcal{N}_G(z_i) = \frac{z_i - \text{mean}_j(z_j)}{\text{std}_j(z_j) + \epsilon}. \quad (3)$$

The coupled Live-SpeechJudge receives one reward: +1 when its overall verdict matches the target and −1 otherwise. Its normalized advantage is broadcast over the full completion. For D-LSJ, external aggregation is $V_w = \text{sgn}[\sum_d w_d(s_d^A - s_d^B)]$, with zero treated as a tie. The aggregate-reward baseline uses $w = (1, 1, 1, 1)$ and broadcasts the resulting single verdict reward over the completion, even though the model emits no conclusion.

Span-local GRPO instead defines, for every confident dimension,

$$r_{i,d} = \begin{cases} +1, & v_{i,d} = \tilde{v}_d, \\ -1, & \text{otherwise,} \end{cases} \quad d \in \{C1, \dots, C4\}, \quad (4)$$

and normalizes each dimension independently across the same rollout group. Its token-level advantage is

$$\hat{A}_{i,t} = \begin{cases} \mathcal{N}_G(r_{i,d}), & t \in T_{i,d} \text{ and } d \text{ is confident,} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Consequently, C1 reward cannot directly cancel C4 reward in the token objective, although all dimensions share model parameters. If every rollout receives the same reward for one dimension, normalization gives zero advantage and that dimension contributes no policy

gradient for the group. Both GRPO variants start from the same two-stage D-LSJ SFT checkpoint and use a new rank-64 LoRA, learning rate 10^{-5} , reverse-KL coefficient 0.04, and three GRPO epochs. We add no format reward.

4.3. Decoupling design and evaluation

D-LSJ reduces cross-dimension coupling in both targets and optimization. Independent teacher queries generate dimension-specific rationales without an overall verdict; pair-dimension masking retains reliable supervision even when other dimensions of the same pair are uncertain. Removing the shared conclusion keeps SFT focused on the four dimension outputs, while span-local GRPO assigns each dimension’s reward only to its own rationale. Together, these mechanisms encourage distinct judgments and dimension-specific credit assignment.

Decoupling does not imply statistical independence: the four dimensions share the same audio, transcript, autoregressive context, and model parameters, and genuine prosodic correlations may remain. We evaluate the resulting judgments in two complementary settings. In the 140-pair multidimensional human set, three annotators assess C1–C4 for every audio pair, with confidence determined separately for each pair-dimension. The larger dimension-specific sets provide additional high-confidence human labels for measuring accuracy on each individual axis.

5. EXPERIMENTS

5.1. Setup and coupled-judge results

Models are trained with SWIFT [16] in bf16 on four 72GB GPUs. All reported training comparisons use one fixed random seed. Gemini uses one judgment in the presented order, reflecting its intended use and the cost of repeated proprietary inference. SpeechJudge-GRM and our students use ten sampled judgments, five in each A/B order. Swapped score pairs are restored to physical clips, each clip’s scores are averaged across the ten generations, and the verdict is the sign of the mean score difference. A tie is counted as an error. This is a stronger inference operating point than single-call Gemini, not a compute-matched comparison.

We report agreement with the final human A/B label. For paired system comparisons, we draw 10,000 bootstrap resamples of test-pair indices and apply identical indices to both systems; the 2.5th and 97.5th percentiles of the accuracy-difference distribution define the 95% interval. These intervals quantify test-pair sampling uncertainty for the fixed checkpoints, not variation across training seeds.

Table 2 shows that LSJ v1+GRPO has higher point accuracy than single-call Gemini in every column. Relative to v1 SFT, its changes are +0.48, +1.68, 0.00, +0.63, and +3.50 points on T1, T1*, T2, T3, and T4. Table 3 reports the paired intervals. Every interval includes zero, so these are descriptive changes. Admitting lower-confidence 3–1 labels (v0) is worse than 4–0-only v1 on four of five columns, indicating that confidence-filtered supervision is more important than simply retaining more teacher-labeled data. Removing the large-gap first stage is associated with a 34.5-point lower T4 estimate, while single-stage training is lower on T2–T4 but higher on T1; these one-seed comparisons do not isolate curriculum order from training dynamics.

On T5, SpeechJudge-GRM obtains 185/222 (83.33%), Gemini obtains 189/222 (85.14%), and LSJ v1+GRPO obtains 192/222 (86.49%). The three-pair, 1.35-point gap over Gemini has no paired

confidence interval and is treated only as descriptive transfer evidence. The comparison is again not compute matched because LSJ uses repeated, balanced-order inference while Gemini uses one call.

5.2. Position-bias diagnostic

We measure position bias using all 4,170 judgments from the 417 T1 pairs under balanced presentation. Each physical comparison occurs in both A/B assignments, so pooling assignments separates presentation slot from clip identity. Let $\Delta_{\text{pos}} = \bar{s}_B - \bar{s}_A$ denote the difference between scores assigned to the second- and first-presented slots. We also report the fraction of raw judgments whose verdict favors presented B; the t statistic is computed from pair-level mean slot differences.

Table 4 shows that SpeechJudge-GRM assigns the second-presented clip nearly one extra point and favors B in two thirds of judgments. In contrast, Live-SpeechJudge is close to slot symmetry in both score gap and win rate. This is consistent with swap-consistency filtering and winner-balanced training order, but it is not a causal ablation of either component. Raw score levels are also not compared across models; the diagnostic concerns within-model A/B asymmetry.

5.3. Dimension-wise evaluation

Across the evaluation sets, D-LSJ produces at least two different non-tie core-dimension verdicts on 24.3–60.6% of pairs, showing that it can avoid unanimous verdict vectors. We additionally use a 140-pair multidimensional human test set. For each audio pair, three trained annotators independently label C1–C4; confidence is determined separately for every pair-dimension, so not every pair has confident labels on all four axes. On this set, the coupled LSJ produces 0/140 non-unanimous core vectors and 48.0% agreement on the confident human dimension labels. D-LSJ SFT produces 90/140 non-unanimous vectors and 67.9% agreement; span-local GRPO changes these to 100/140 (71.4%) and 71.4%, respectively. This same-set comparison evaluates both resistance to verdict collapse and agreement with the retained human judgments.

Table 5 therefore evaluates each dimension against a separate, unanimous human test set. Ten-sample SFT exceeds one-call Gemini in every dimension. Span-local GRPO changes pooled agreement from 73.64% to 77.94% with one sample and from 84.10% to 86.10% with ten samples. At ten samples it is 1.00 point above aggregate-reward GRPO. Table 6 shows that all per-dimension paired intervals for span-local GRPO versus SFT and aggregate-reward GRPO include zero; we therefore report point estimates without claiming statistical significance.

5.4. Additional diagnostics

Repeated, balanced-order inference contributes unevenly. Single-sample LSJ v1+GRPO obtains 67.87, 83.19, 65.77, 80.95, and 74.50% on T1, T1*, T2, T3, and T4; ten-sample aggregation changes these estimates by +3.35, −0.84, +4.50, +1.91, and +9.00 points. It combines repeated generation with order balancing, whose effects are not isolated here.

D-LSJ also permits external scalar decisions without restoring an overall text target. Validation-selected weights $w = (3, 0.5, 0.5, 3)$ give 73.6% on T1 and 85.5% on T4, versus 69.3% and 78.5% with uniform weights; T2 remains 71.2%. This is not a causal comparison with the coupled judge because their objectives differ.

We also test CEAEval’s adaptive audio-attention bias [13]. A representative instance contains 426 audio, 634 rationale, and 2,618

Table 2. Overall human-label agreement (%). “1s” denotes the first sampled judgment and “10s” denotes 10-sample balanced-order aggregation; Gemini uses one call. T1★ is part of T1.

Model	T1	T1★	T2	T3	T4
SpeechJudge-GRM (10s)	55.88	72.27	48.65	53.97	42.00
Gemini-3.1-pro-preview (1s)	65.23	72.27	65.77	73.33	58.00
Without stage-1 SFT (10s)	53.00	72.27	60.36	69.84	45.50
Single-stage SFT (10s)	72.66	80.67	67.57	80.00	74.50
LSJ v0, 4–0 + 3–1 (10s)	71.22	78.99	65.77	80.63	73.00
LSJ v1, 4–0 SFT (1s)	64.99	74.79	63.06	73.97	66.50
LSJ v1, 4–0 SFT (10s)	70.74	80.67	70.27	82.22	80.00
LSJ v1 + GRPO (1s)	67.87	83.19	65.77	80.95	74.50
LSJ v1 + GRPO (10s)	71.22	82.35	70.27	82.86	83.50

Table 3. Paired-bootstrap comparison of 10-sample LSJ v1+GRPO against v1 SFT. Differences and 95% intervals are in percentage points.

Set	GRPO – SFT	95% interval
T1	+0.48	[−2.64, +3.60]
T1★	+1.68	[−2.52, +6.72]
T2	0.00	[−7.21, +7.21]
T3	+0.63	[−1.90, +3.18]
T4	+3.50	[−1.50, +8.50]

Table 4. Position statistics on T1 under 10-sample balanced-order inference.

Model	$\bar{s}_A$	$\bar{s}_B$	Δ_{pos}	t	B win
Live-SpeechJudge	7.842	7.854	+0.012	+0.4	50.7%
SpeechJudge-GRM	7.061	8.039	+0.978	+11.7	66.8%

rubric-prompt tokens. It trains stably but does not improve held-out accuracy, so it is excluded from the final model. One possible explanation is that the generated rationale is not much longer than the audio sequence, limiting the benefit of explicitly up-weighting audio attention; substantially longer generations may behave differently.

5.5. Best-of-8 and limitations

For each transcript, the TTS model generates eight candidates and Live-SpeechJudge evaluates all $\binom{8}{2} = 28$ distinct pairs. Without requiring absolute-score calibration, we select the candidate with the largest number of pairwise wins,

$$\hat{a} = \arg \max_{a_i} \sum_{j \neq i} \mathbf{1}[V(a_i, a_j) = a_i]. \quad (6)$$

Two trained contractor annotators independently rank the top three candidates for 400 sets. We retain the 136 sets for which both produce the same ordered top-three ranking; this inclusion decision is made without model predictions. On this high-confidence subset, the selector obtains Hit@1/2/3 of 98/136 (72.06%), 106/136 (77.94%), and 116/136 (85.29%), compared with random-selection baselines of 12.5%, 25.0%, and 37.5%. Hit@ k means that the tournament winner lies within the agreed human top- k . These rates characterize the retained subset and do not estimate performance on all 400 collected sets.

Table 5. D-LSJ agreement with unanimous human dimension labels (%). “Agg.” and “Span” denote aggregate-reward and span-local GRPO.

Model	C1	C2	C3	C4	All
Gemini (1s)	74.55	77.33	80.28	84.38	78.65
SFT (1s)	68.75	65.70	76.06	86.88	73.64
SFT (10s)	79.02	80.81	85.92	93.13	84.10
Agg. GRPO (1s)	70.54	71.51	79.58	90.62	77.22
Agg. GRPO (10s)	79.91	83.14	87.32	92.50	85.10
Span GRPO (1s)	71.88	76.74	76.76	88.75	77.94
Span GRPO (10s)	83.04	83.14	85.21	94.38	86.10

Table 6. Ten-sample span-local GRPO differences and paired-bootstrap 95% intervals, in percentage points.

Dim.	Versus SFT	Versus Agg. GRPO
C1	+4.02 [−0.89, +9.38]	+3.12 [−1.79, +7.59]
C2	+2.33 [−2.33, +6.98]	0.00 [−5.23, +5.23]
C3	−0.70 [−4.23, +2.82]	−2.11 [−6.34, +2.11]
C4	+1.25 [−1.88, +4.38]	+1.88 [−1.88, +5.62]

Three limitations bound the evidence. First, training results use one seed; bootstrap intervals quantify test-pair sampling uncertainty, not variation across training runs. Second, ten-sample comparisons against Gemini use more inference and balanced order. Third, the human dimension sets evaluate one target dimension per pair rather than complete human C1–C4 vectors. D-LSJ is currently limited to C1–C4, and using its vector rewards to optimize a TTS model remains future work.

6. CONCLUSION

We presented Live-SpeechJudge and D-LSJ for multi-dimensional live streaming prosody evaluation. Balanced-order Live-SpeechJudge achieves strong point accuracy, shows substantially less measured slot bias than SpeechJudge-GRM, and supports Best-of-8 selection. D-LSJ removes the shared overall-verdict target and localizes SFT and GRPO supervision by dimension, producing varied judgments and 86.10% pooled agreement. Its per-dimension improvements remain descriptive because all confidence intervals include zero.

7. ACKNOWLEDGMENTS

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8. COMPLIANCE WITH ETHICAL STANDARDS

Human labels were produced by contracted annotation staff who received task-specific training and were blinded to system identity. We report only aggregate results. Evaluation audio is not publicly redistributed because source recordings are subject to copyright and licensing restrictions.

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